Modeling Conditional Gene Dependencies in Breast Cancer through Sparse Precision Matrix Estimation

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1 Abstract

Estimating sparse inverse covariance matrices is central to understanding conditional dependence structures among high-dimensional variables. We study the Constrained ℓ_1 Minimization for Inverse Matrix Estimation (CLIME) method for estimating a sparse precision matrix from an i.i.d. sample. The estimator recovers the underlying graphical structure by promoting sparsity in the precision matrix, where non-zeros correspond to conditional dependencies. Our primary objective in this report is to apply CLIME to a real-data example. Specifically, we analyze a breast cancer gene-expression dataset from Hess et al. (2006), available at `http://bioinformatics.mdanderson.org/`. We use CLIME to learn the gene–gene conditional dependency network via the inverse covariance matrix and then perform linear discriminant analysis (LDA) to predict clinical response. We also assess numerical performance on simulated datasets generated from both sparse and dense precision matrices, benchmarking CLIME against established alternatives including the Graphical Lasso (Glasso), nodewise regression, and SCAD-type nonconvex penalization. The comparison highlights that CLIME operates under notably weaker structural conditions on the population precision matrix while achieving convergence rates that match or, in some regimes, surpass competing methods. Its linear-programming formulation further reduces computational burden, making it especially attractive in high-dimensional applications.

2 Introduction

Let $X = (X_1, \ldots, X_p)^\top$ be a p-variate random vector with covariance matrix Σ_0 and precision matrix $\Omega_0 := \Sigma_0^{-1}$. Given an i.i.d. sample $\{X_1, \ldots, X_n\}$ from the distribution of X, the natural estimator of Σ_0 is the sample covariance

$$\hat{\Sigma}_n = \frac{1}{n}\sum_{k=1}^{n}(X_k - \bar{X})(X_k - \bar{X})^\top, \qquad \bar{X} = \frac{1}{n}\sum_{k=1}^{n} X_k.$$

However, $\hat{\Sigma}_n$ is singular when $p > n$, so its inverse is not a viable estimator of Ω_0. To achieve consistent estimation in high dimensions, one typically imposes structure (e.g., sparsity or low rank) and uses regularization.

A closely related goal is support recovery of the precision matrix, which corresponds to graphical model selection. We use the same notations used in the class. Let $G = (V, E)$ be an undirected graph encoding conditional independence among the components of X. The vertex set V has p nodes corresponding to $X_1, \ldots, X_p$, and the edge set E contains unordered pairs (i, j) indicating edges. In a Gaussian model $X \sim \mathcal{N}(\mu_0, \Sigma_0)$ with precision $\Omega_0 = (\omega_{ij}^0)$, we have $\omega_{ij}^0 = 0$ if and only if $X_i \perp X_j \mid \{X_k : k \neq i, j\}$. Thus, recovering the edges of the graph amounts to identifying the zero–nonzero pattern of Ω_0. Estimation of the precision matrix Ω_0 is more involved due to the lack of a natural pivotal estimator like $\hat{\Sigma}_n$. Assuming certain ordering structures, methods based on banding the Cholesky factor of the inverse have been proposed and studied; see, e.g., Bickel and Levina (2008). Penalized likelihood methods have also been introduced for estimating sparse precision matrices. In particular, the ℓ_1-penalized normal likelihood estimator and its variants, which we refer to as ℓ_1-MLE type estimators, have been considered in several papers; see, for example, Yuan and Lin (2007), Friedman et al. (2008), and Rothman et al. (2008).

To set the motivation for CLIME (introduced by Tony Cai), we briefly review the Graphical Lasso (Glasso), based on the ℓ_1-regularized log-determinant program:

$$\hat{\Omega}^{\text{Glasso}} := \arg\min_{\Omega \succ 0} \left\{ \operatorname{tr}(\hat{\Sigma}_n \Omega) - \log\det(\Omega) + \lambda_n \|\Omega\|_1 \right\}, \tag{1}$$

where $\hat{\Sigma}_n$ is the sample covariance and $\lambda_n > 0$ is a tuning parameter. The optimal solution $\hat{\Omega}^{\text{Glasso}}$ satisfies the KKT condition

$$(\hat{\Omega}^{\text{Glasso}})^{-1} - \hat{\Sigma}_n = \lambda_n \hat{Z}, \tag{2}$$

where $\hat{Z} \in \partial \|\hat{\Omega}^{\text{Glasso}}\|_1$ is a subgradient of the ℓ_1-norm at $\hat{\Omega}^{\text{Glasso}}$. This motivates the constrained ℓ_1-minimization

$$\min_{\Omega \in \mathbb{R}^{p\times p}} \|\Omega\|_1 \quad \text{subject to} \quad \left\|\Omega^{-1} - \hat{\Sigma}_n\right\|_\infty \le \lambda_n. \tag{3}$$

Because this constraint is difficult to handle directly, one can multiply the constraint by Ω and relax it to obtain a tractable convex program—the basis of the CLIME estimator. The authors define the CLIME estimator as follows. Let $\hat{\Omega}^{(1)}$ be a solution of

$$\hat{\Omega}^{(1)} \in \arg\min_{\Omega \in \mathbb{R}^{p\times p}} \|\Omega\|_1 \quad \text{subject to} \quad \|\hat{\Sigma}_n \Omega - I_p\|_\infty \le \lambda_n, \tag{4}$$

where $\hat{\Sigma}_n$ is the sample covariance, $\lambda_n > 0$ is a tuning parameter, and $\|\cdot\|_1$, $\|\cdot\|_\infty$ denote the elementwise ℓ_1- and ℓ_∞-norms, respectively. In (4), no symmetry constraint is imposed, so $\hat{\Omega}^{(1)}$ need not be symmetric.

$$\hat{\Omega}^{(1)} = \left(\hat{\omega}_{ij}^{(1)}\right)_{1\le i,j\le p} = \left(\hat{\omega}_1^{(1)}, \ldots, \hat{\omega}_p^{(1)}\right),$$

where $\hat{\omega}_j^{(1)}$ denotes the jth column of $\hat{\Omega}^{(1)}$. The final CLIME estimator $\hat{\Omega}$ is obtained by symmetrization:

$$\hat{\Omega} = (\hat{\omega}_{ij})_{1\le i,j\le p}, \qquad \hat{\omega}_{ij} = \hat{\omega}_{ji} = \hat{\omega}_{ij}^{(1)} \mathbf{1}\{|\hat{\omega}_{ij}^{(1)}| \le |\hat{\omega}_{ji}^{(1)}|\} + \hat{\omega}_{ji}^{(1)} \mathbf{1}\{|\hat{\omega}_{ij}^{(1)}| > |\hat{\omega}_{ji}^{(1)}|\}.$$

Thus for each pair (i,j) we retain the entry with smaller magnitude between $\hat{\omega}_{ij}^{(1)}$ and $\hat{\omega}_{ji}^{(1)}$. Clearly, $\hat{\Omega}$ is symmetric, and Theorem 1 shows it is positive definite with high probability. One benefit is that the convex program (4) decomposes into p vector problems. Let e_i denote the ith standard basis vector in $\mathbb{R}^p$. For $1 \le i \le p$, define $\hat{\beta}_i$ as the solution of

$$\hat{\beta}_i \in \arg\min_{\beta \in \mathbb{R}^p} \|\beta\|_1 \quad \text{subject to} \quad \|\hat{\Sigma}_n \beta - e_i\|_\infty \le \lambda_n. \tag{5}$$

Let $\hat{B} = (\hat{\beta}_1, \ldots, \hat{\beta}_p)$. Then the solution set of (4) coincides with $\{\hat{B}\}$; that is,

$$\{\hat{\Omega}^{(1)}\} = \{\hat{B}\}.$$

This equivalence is useful for implementation (columnwise parallelization) and for analysis.

3 Computational Benefits

Since (4) and (5) are equivalent via a columnwise decomposition, (5) reduces to a linear program:

$$\min_{\beta\in\mathbb{R}^p} \|\beta\|_1 \quad \text{s.t.} \quad \|\hat{\Sigma}_n\beta - e_i\|_\infty \le \lambda_n.$$

Relaxation via auxiliary variables: introduce $u_j \ge 0$ so that $|\beta_j| \le u_j$. Then

$$\min_{\beta,u} \sum_{j=1}^{p} u_j$$

subject to

$$-u_j \le \beta_j \le u_j \quad \text{for } j = 1, \dots, p,$$

and

$$\|\hat{\Sigma}_n\beta - e_i\|_\infty \le \lambda_n \quad \iff \quad -\lambda_n \le \hat{\sigma}_k^{\mathsf{T}}\beta - \mathbf{1}\{k = i\} \le \lambda_n \ \text{ for } k = 1, \dots, p,$$

which is equivalent to the pair of linear inequalities

$$-\hat{\sigma}_k^{\mathsf{T}}\beta - \mathbf{1}\{k = i\} \le \lambda_n, \quad \hat{\sigma}_k^{\mathsf{T}}\beta - \mathbf{1}\{k = i\} \le \lambda_n. \tag{6}$$

These constraints define a polytope in $\mathbb{R}^p$, and the objective is linear in (β, u), so each column subproblem is a standard linear program. Because the problem is separable across columns, the p subproblems can be solved in parallel. In practice, the overall computational cost scales favorably with p using modern LP solvers.

4 Theoretical Guarantees

From a statistical perspective, CLIME minimizes the ℓ_1 norm of the precision matrix subject to constraints derived from the sample covariance. We establish convergence rates under multiple norms: the spectral norm, Frobenius norm, and elementwise ℓ_∞ norm. Let

$$\widehat{\Sigma} = (\widehat{\sigma}_{ij}) \in \mathbb{R}^{p\times p}, \qquad \Sigma^0 = (\sigma_{ij}^0) \in \mathbb{R}^{p\times p}, \qquad \mathbb{E}X = \mu = (\mu_1, \dots, \mu_p)^\top.$$

It is conventional to analyze performance under different tail conditions on X.
Assumptions.

(C1) Exponential-type tails: There exists $0 < \eta < 1/4$ such that $\log p/n \le \eta$ and

$$\mathbb{E}\exp\left\{t\,(X_i - \mu_i)^2\right\} \le K < \infty \quad \text{for all } |t| \le \eta \text{ and all } i,$$

for a finite constant K.

(C2) Polynomial-type tails: For some $\gamma, c_1 > 0$, $p \le c_1 n^\gamma$, and for some $\delta > 0$,

$$\mathbb{E}\left|X_i - \mu_i\right|^{4\gamma+4+\delta} \le K \quad \text{for all } i.$$

The authors define the parameter space

$$\mathcal{U}\big(q, s_0(p)\big) = \Big\{\Omega = (\omega_{ij}) \in \mathbb{R}^{p\times p}: \ \|\Omega\|_{L_1} \le M, \quad \max_{1\le i\le p} \sum_{j=1}^{p} |\omega_{ij}|^{\,q} \le s_0(p)\Big\}, \qquad 0 \le q < 1,$$

where $\|\Omega\|_{L_1} := \max_{1\le j\le p} \sum_{i=1}^{p} |\omega_{ij}|$ and $M > 0$ is fixed. When $q = 0$, $\mathcal{U}(0, s_0(p))$ corresponds to the class of $s_0(p)$-sparse matrices. Let

$$\theta \;=\; \max_{i,j}\; \mathbb{E}\Big[\,(X_i - \mu_i)(X_j - \mu_j) - \sigma_{ij}^0\,\Big]^2 \;=:\; \max_{i,j}\; \theta_{ij},$$

so θ_{ij} relates to the variance of $\widehat{\sigma}_{ij}$, and θ captures the overall variability of $\widehat{\Sigma}$. Under either (**C1**) or (**C2**), θ is bounded by a constant depending only on γ, δ, K.

Suppose $\Omega_0 \in \mathcal{U}\big(q, s_0(p)\big)$.

(a) Under (C1), let $\lambda_n = C_0\, M \sqrt{\frac{\log p}{n}}$, where

$$C_0 \;=\; 2\,\eta^{-2}\left(2+\tau+\eta^{-1}e^{2K^2}\right)^2, \qquad \tau > 0.$$

Then

$$\big\|\widehat{\Omega} - \Omega_0\big\|_2 \;\le\; C_1\, M^{\,2-2q}\, s_0(p) \left(\frac{\log p}{n}\right)^{\frac{1-q}{2}}, \tag{7}$$

with probability at least $1 - 4p^{-\tau}$, where

$$C_1 \;\le\; 2\left(1 + 2^{\,1-q} + 3^{\,1-q}\right) 4^{\,1-q}\, C_0^{1-q}.$$

(b) Under (C2), let $\lambda_n = C_2\, M \sqrt{\frac{\log p}{n}}$, where

$$C_2 \;=\; \sqrt{(5+\tau)\,(\theta+1)}.$$

Then

$$\big\|\widehat{\Omega} - \Omega_0\big\|_2 \;\le\; C_3\, M^{\,2-2q}\, s_0(p) \left(\frac{\log p}{n}\right)^{\frac{1-q}{2}}, \tag{8}$$

with probability at least $1 - O\big(n^{-\delta/8} + p^{-\tau/2}\big)$, where

$$C_3 \;\le\; 2\left(1 + 2^{\,1-q} + 3^{\,1-q}\right) 4^{\,1-q}\, C_2^{1-q}.$$

When M is independent of n and p, the dependence on (n, p) is fully captured by the factor $(\log p/n)^{(1-q)/2}$.

Comparison with Graphical Lasso

- Performance: In the polynomial-tail case with $q = 0$, the rate in (8) is substantially better than the rate $O\Big(s_0(p)\, \frac{p^{1/(\gamma+1+\delta/4)}}{n}\Big)$ for the ℓ_1-regularized MLE reported by Ravikumar et al. (2009).
- Assumptions: Glasso typically requires exact sparsity and strong incoherence conditions to guarantee ℓ_2-error control assumptions that are difficult to verify in practice. CLIME avoids these stringent requirements.

Incoherence / irrepresentability: Let $\Sigma^\star = (\Omega^\star)^{-1}$ and $\Gamma^\star = \Sigma^\star \otimes \Sigma^\star$. Partition by support S vs S^c:

$$\big\|\Gamma^\star_{S^cS}(\Gamma^\star_{SS})^{-1}\big\|_\infty \;\le\; 1-\alpha, \qquad \alpha \in (0,1).$$

Intuitively, a zero entry of $\Omega^\star$ cannot be too strongly correlated (in the Fisher-information sense) with the collection of nonzero entries. Formally, every row of $\Gamma^\star_{S^cS}(\Gamma^\star_{SS})^{-1}$ must have ℓ_1-norm at most $1-\alpha$, i.e., strictly bounded away from 1. This requirement can limit the applicability of Glasso in regimes with complex correlation structures, even when the precision matrix is sparse.

5 Competing Benchmarks

We compare CLIME to several common alternatives.

1. Nodewise Lasso regressions. For each $j \in \{1, \dots, p\}$, solve

$$\widehat{\beta}^{(j)} \in \arg\min_{\beta \in \mathbb{R}^{p-1}} \left\{ \frac{1}{2n} \|X_j - X_{-j}\beta\|_2^2 + \lambda_j \|\beta\|_1 \right\},$$

   with $\lambda_j \asymp \sqrt{(\log p)/n}$.

   Residual variance and assembly. Let $\widehat{\sigma}_j^2 = (1/n)\,\|X_j - X_{-j}\widehat{\beta}^{(j)}\|_2^2$. Define

$$\widehat{\Omega}_{jj} = \frac{1}{\widehat{\sigma}_j^2}, \qquad \widehat{\Omega}_{ij} = -\,\widehat{\Omega}_{jj}\,\widehat{\beta}_i^{(j)}, \quad i \neq j,$$

   using the identities $\Omega^\star_{ij} = -\Omega^\star_{jj}\beta^\star_{ij}$ and $\Omega^\star_{jj} = 1/\operatorname{Var}(X_j \mid X_{-j})$.

2. SCAD-penalized estimator:

$$\hat{\Omega}_{\text{SCAD}} := \arg\min_{\Omega \succ 0} \left\{ \langle \Omega, \Sigma_n \rangle - \log\det(\Omega) + \sum_{i=1}^{p} \sum_{j=1}^{p} \text{SCAD}_{\lambda,a}(|\omega_{ij}|) \right\}.$$

   Let $t \geq 0$, $\lambda > 0$, $a > 2$. The derivative of the SCAD penalty is

$$\text{SCAD}_{\lambda,a}(t) = \begin{cases} \lambda, & 0 \leq t \leq \lambda, \\ \dfrac{a\lambda - t}{a-1}, & \lambda < t \leq a\lambda, \\ 0, & t > a\lambda. \end{cases}$$

   SCAD employs a nonconvex penalty that reduces bias relative to the ℓ_1 penalty, but it is typically more computationally demanding.

6 Simulation Studies

We compare the numerical performance of four estimators of the precision matrix: the CLIME estimator ($\widehat{\Omega}_{\text{CLIME}}$), Graphical Lasso ($\widehat{\Omega}_{\text{Glasso}}$), nodewise regression estimator ($\widehat{\Omega}_{\text{nodewise}}$), and the SCAD estimator ($\widehat{\Omega}_{\text{SCAD}}$). We consider the following models for the true precision matrix $\Omega_0 \in \mathbb{R}^{p \times p}$:

1. Model 1 (banded Toeplitz): $\omega^0_{ij} = 0.6^{\,|i-j|}$. This yields a banded structure with entries decaying away from the diagonal.

2. Model 2 (dense): Ω_0 has all off-diagonal elements equal to 0.5 and diagonal elements equal to 1. This serves as a nonsparse example.

3. Model 3 (Rothman et al., 2008). Let $\Omega_0 = B + \delta I_p$, where B is symmetric with $B_{ii} = 0$ and, for $i < j$, each off-diagonal entry is generated independently as $B_{ij} \in \{0.5, 0\}$ with $\Pr(B_{ij} = 0.5) = 0.1$ and $\Pr(B_{ij} = 0) = 0.9$. Choose δ so that the condition number $\kappa(\Omega_0) = \sigma_{\max}(\Omega_0)/\sigma_{\min}(\Omega_0)$ equals p. Finally, standardize to unit diagonals by setting $D = \operatorname{diag}\big(1/\sqrt{\Omega_{0,ii}}\big)$ and replacing Ω_0 with $D\,\Omega_0\,D$, so that $\operatorname{diag}(\Omega_0) = \mathbf{1}$.

For each model, we generate a training sample of size $n = 100$ from a p-variate normal distribution with mean zero and covariance matrix $\Sigma_0 = \Omega_0^{-1}$, and an independent validation sample of size $n = 100$ from the same distribution. Using the training data, we compute each estimator after cross-validating λ over a grid 1000 equally spaced (in log scale) tuning values λ and select the one that minimizes the validation likelihood loss

$$L(\Omega;\,\widehat{\Sigma}) \;=\; \operatorname{tr}\big(\widehat{\Sigma}\,\Omega\big) \;-\; \log\det(\Omega),$$

where $\widehat{\Sigma}$ is the sample covariance computed from the validation set. Note that since CLIME is a columnwise separable linear problem, we need to break down the optimization problem in p smaller sub-problems. For each predictor, we now calculate minimum value of $\|\Omega\|_1$ in $\mathcal{O}(p^2)$ many points. Hence, the total computational complexity for the CLIME problem is only $\mathcal{O}(p^3)$. The Glasso and SCAD estimators are fitted on the same training/testing splits using the same cross-validation scheme. We consider dimensions $p \in \{20, 30, 40, 50, 100\}$ and replicate each experiment 100 times.

Estimation quality is measured by l_2 norm. The averages of these losses (across replications) are reported in Table 1, 2, and 3. Overall, CLIME nearly uniformly outperforms Glasso; the improvement is slightly more pronounced for sparse models as p grows, though not significant in absolute terms. SCAD has a non-convex penalty and is the most computationally costly among all, but achieves similar accuracy when p is large. Note that SCAD employs a nonconvex penalty to reduce bias, whereas CLIME optimizes a convex ℓ_1 objective efficiently.

	Model 1: $\Omega^0_{ij} = 0.6^{\lvert i-j \rvert}$			
p	$\hat{\Omega}_{\text{CLIME}}$	$\hat{\Omega}_{\text{Glasso}}$	$\hat{\Omega}_{\text{Nodewise}}$	$\hat{\Omega}_{\text{SCAD}}$
20	0.114	0.121	0.122	0.108
30	0.076	0.082	0.081	0.102
40	0.041	0.040	0.045	0.045
50	0.031	0.032	0.029	0.032
100	0.025	0.023	0.027	0.024

Table 1: Ratio of l_2 loss between true and predicted precision matrix and the l_2 norm of the true precision. CLIME performance is comparable with the existing methods with significantly less computational complexity.

	Model 2: $\Omega^0_{ij} = 0.5 \ \ (i \neq j), \ \ \Omega^0_{ii} = 1$			
p	$\hat{\Omega}_{\text{CLIME}}$	$\hat{\Omega}_{\text{Glasso}}$	$\hat{\Omega}_{\text{Nodewise}}$	$\hat{\Omega}_{\text{SCAD}}$
20	0.871	0.912	0.451	1.028
30	1.653	1.423	1.022	1.836
40	1.126	0.920	0.721	1.741
50	1.237	0.993	0.821	1.823
100	1.172	0.943	0.692	1.793

Table 2: Similar ratio. Since the problem is not sparse anymore, no method should perform well. Nodewise regression and Glasso outperforms CLIME.

	Model 3			
p	$\hat{\Omega}_{\text{CLIME}}$	$\hat{\Omega}_{\text{Glasso}}$	$\hat{\Omega}_{\text{Nodewise}}$	$\hat{\Omega}_{\text{SCAD}}$
20	1.601	1.729	1.922	1.384
30	1.622	1.209	1.208	1.160
40	1.203	1.001	0.991	0.792
50	1.237	0.993	0.821	0.532
100	0.967	0.897	0.819	0.320

Table 3: Again a sparse problem. In the l_2 norm sense, SCAD performs a little better than others.

Now we plot the true and estimated precision matrices to verify which methods identifies the gene dependencies the best. In Figure 1 and 2, the true precision matrix is

$$\omega^0_{ij} = 0.6^{\,\lvert i-j \rvert}$$

As we can see, the precision matrix is not exactly sparse but nearly/limiting sparse. Every off-diagonal is nonzero, so exact sparsity is violated ($s_0(p) = p$), hence graphical lasso performs poorly. However, CLIME assumes approximate sparsity via

$$\Omega_0 \in \mathcal{U}\big(q, s_0(p)\big), \qquad \max_{1\le i\le p} \sum_{j=1}^{p} |\omega_{ij}|^{\,q} \le s_0(p), \qquad 0 \le q < 1.$$

In this model,

$$\sum_{j=1}^{p} \big|\Omega^0_{ij}\big|^{\,q} \;=\; \sum_{k=-\infty}^{\infty} 0.6^{\,q|k|}$$

is a geometric series, so it is bounded by a constant independent of p for any fixed $q > 0$. Hence, CLIME works better.

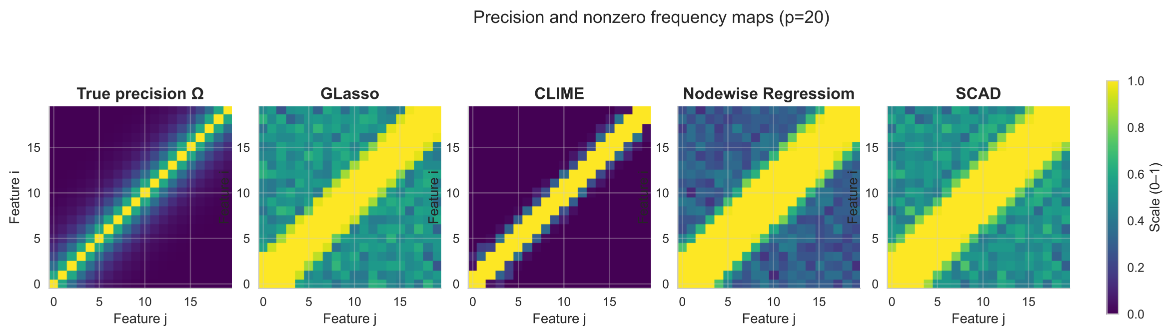

Figure 1: Estimated precision matrix for different estimators under $n = 100, p = 20$

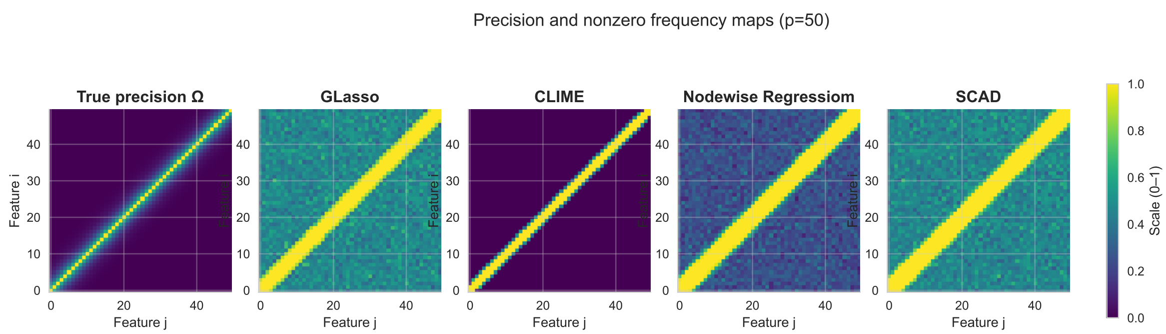

Figure 2: Estimated precision matrix for different estimators under $n = 100, p = 50$. As described above, CLIME performance is the best.

For model 2,

$$\Omega^0_{ij} = \begin{cases} 1, & i = j, \\ 0.5, & i \ne j, \end{cases} \qquad 1 \le i, j \le p.$$

There is no sparsity in the precision matrix at all. Ideally, no estimation method should work, nodewise regression outperforms CLIME in this case, see Figure 3.

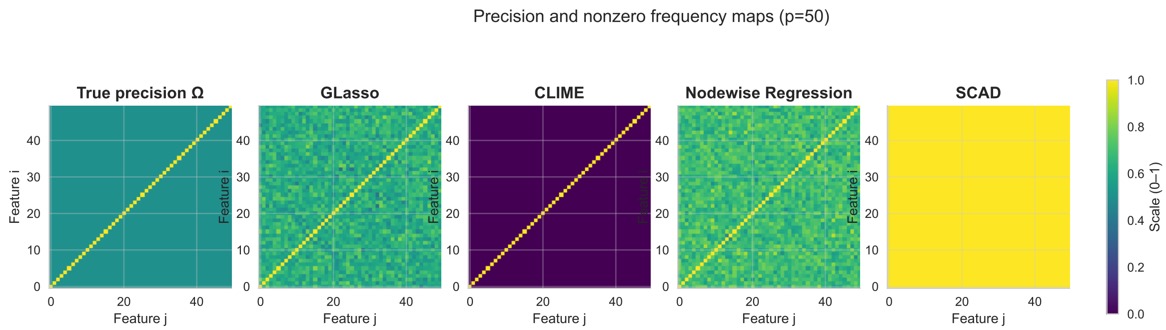

Figure 3: The precision matrix is dense. Exact sparsity fails, limiting sparsity (for CLIME) fails as well. Nodewise regression outperforms.

Model 3 is a good combination of sparsity and dense precision. There are a few off-diagonal entries having significant correlation (see Figure 4). Although l_2 loss comparison implies SCAD works the best (see Table 3), from the perspective of identifying an edge between the edges (covariate dependency), CLIME outperforms.

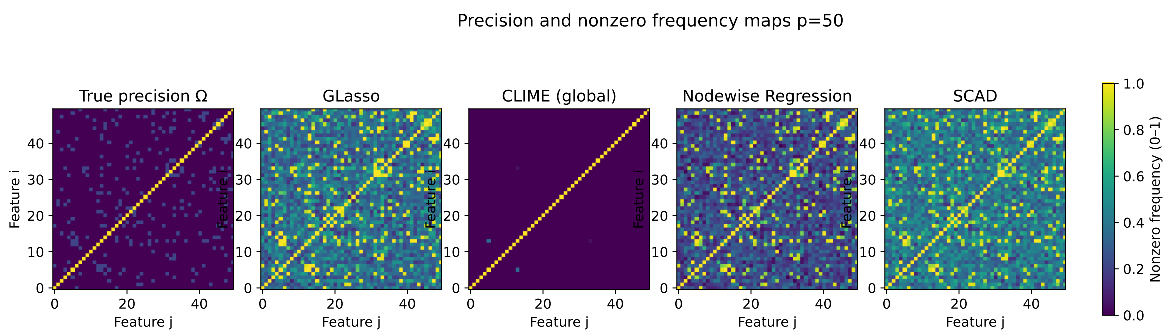

Figure 4: Estimated precision matrix for different estimators under $n = 100, p = 50$

Analysis of the breast cancer dataset

We now apply the CLIME method to a real data example. The breast cancer data analyzed by Hess et al. (2006), available at `http://bioinformatics.mdanderson.org/`. We could not find the exact data used in the paper, but an updated dataset more number of subjects. Our dataset consists of 22,283 gene expression measurements on 508 subjects, including 105 subjects with pathological complete response (pCR) and 403 subjects with residual disease (RD). Because pCR is associated with a high chance of long-term cancer-free survival, it is of particular interest to study patient response (pCR or RD). Based on the estimated inverse covariance matrix of the gene expression levels, we apply linear discriminant analysis (LDA) to predict whether a subject attains the pCR state. For a fair comparison with other inverse covariance estimators, we follow the analysis scheme in Cai et al. (2011). For completeness, we summarize the steps:

- The data are randomly split into a training set and a testing set using stratified sampling: 25 pCR subjects and 80 RD subjects are randomly assigned to the test set (roughly one sixth of each group), and the remaining subjects form the training set.

- On the training set, a two-sample t-test is performed gene-by-gene between the two groups; the

500 most significant genes (smallest p-values) are retained as covariates for prediction. Note the training sample size is 403, less than the covariate dimension, so this examines the $p > n$ regime.

- The selected gene data are standardized using the sample standard deviations estimated from the training set.
- Under the LDA framework, the normalized gene expression vectors are modeled as $X \sim \mathcal{N}(\mu_k, \Sigma)$, where both groups share the same covariance matrix Σ but have different means μ_k, $k = 1$ (pCR) and $k = 2$ (RD). The inverse covariance estimate $\widehat{\Omega}$ from each method is then used in the LDA scores

$$\delta_k(x) \ = \ x^\top \widehat{\Omega}\, \widehat{\mu}_k \ - \ \frac{1}{2}\, \widehat{\mu}_k^\top \widehat{\Omega}\, \widehat{\mu}_k \ + \ \log \widehat{\pi}_k, \tag{9}$$

  where $\widehat{\pi}_k = n_k/n$ is the proportion of group-k subjects in the training set and $\widehat{\mu}_k = \frac{1}{n_k} \sum_{i \in \text{group } k} x_i$ is the within-group sample mean in the training set.
- The classification rule is $\widehat{k}(x) = \arg\max_{k \in \{1,2\}} \delta_k(x)$.
- We then repeat this process 100 times for cross-validation and take the average of sensitivity, specificity, and Matthews correlation coefficient (MCC) as a measure for model performance.

This estimation and classification performance is the same criteria as Fan et al. (2009). for comparability. Tuning parameters are selected via cross-validation on the training data, and the entire procedure is repeated 100 times. Specificity, sensitivity, Matthews correlation coefficient (MCC), defined as:

$$\text{Specificity} \ = \ \frac{\text{TN}}{\text{TN} + \text{FP}}, \qquad \text{Sensitivity} \ = \ \frac{\text{TP}}{\text{TP} + \text{FN}},$$

$$\text{MCC} \ = \ \frac{\text{TP} \times \text{TN} - \text{FP} \times \text{FN}}{\sqrt{(\text{TP} + \text{FP})(\text{TP} + \text{FN})(\text{TN} + \text{FP})(\text{TN} + \text{FN})}},$$

where TP/TN denote true positives (pCR) and true negatives (RD), and FP/FN denote false positives/negatives. Larger values indicate better performance. The averages of these criteria are listed over 100 replications, are reported in Table 4. CLIME markedly improves sensitivity and is comparable in specificity. Overall performance, measured by MCC, overwhelmingly favors CLIME. CLIME also yields the sparsest precision matrix, which is typically advantageous for interpretation on real datasets.

Method	Sensitivity	Specificity	MCC
Glasso	0.674	0.767	0.403
Nodewise	0.721	0.792	0.477
SCAD	0.612	0.820	0.492
CLIME	0.829	0.783	0.505

Table 4: Performance of different precision matrix estimators in breast cancer dataset. CLIME has the best predictive power in terms of identifying the true positive even in an imbalanced setup. The specificity performance of SCAD is **spurious**, it assigns every subject to RD, and because of data imbalance (higher number of RD population) we see the number blown up.

The left block of Figure 5 shows the true covariance matrix, while the right one is the true precision matrix for the 50 most important genes in this data. We permute the columns for a better visual representation. The covariance matrix indicates a significant correlation among the genes which are closer to each other is this representation, as they start getting away the dependency fades away. There are a very few significant number of off-diagonal entries in the precision matrix.

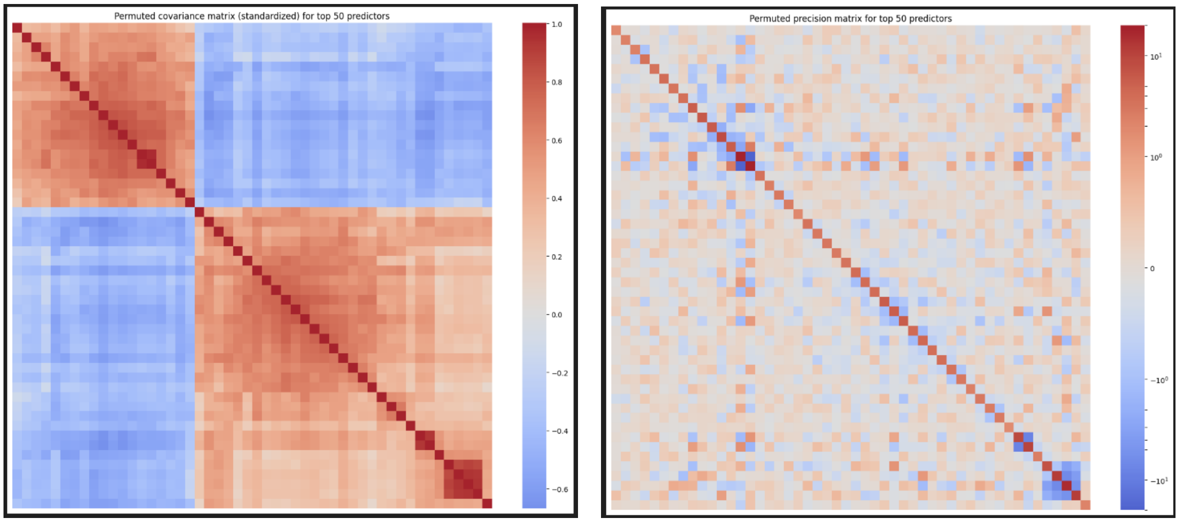

Figure 5: The left panel is the true covariance matrix and the right panel is the inverse matrix (only 50 most important predictors considered). The off-diagonal entries of the true precision matrix are not exactly zero, however, very close to zero. This is a perfect setting for CLIME to perform well.

Next, in Figure 6, we plot the estimated precision matrix by Glasso and CLIME. We can see since the off-diagonal entries of the true precision matrix are not exactly zero, exact sparsity fails and estimated precision by Glasso is poor. However, the limiting-sparsity assumption for CLIME still holds and CLIME outperforms. Thus, we can now scientifically justify why CLIME performs better in this context. Because of the limiting-sparsity nature of the true precision matrix, CLIME can identify the gene dependencies much better.

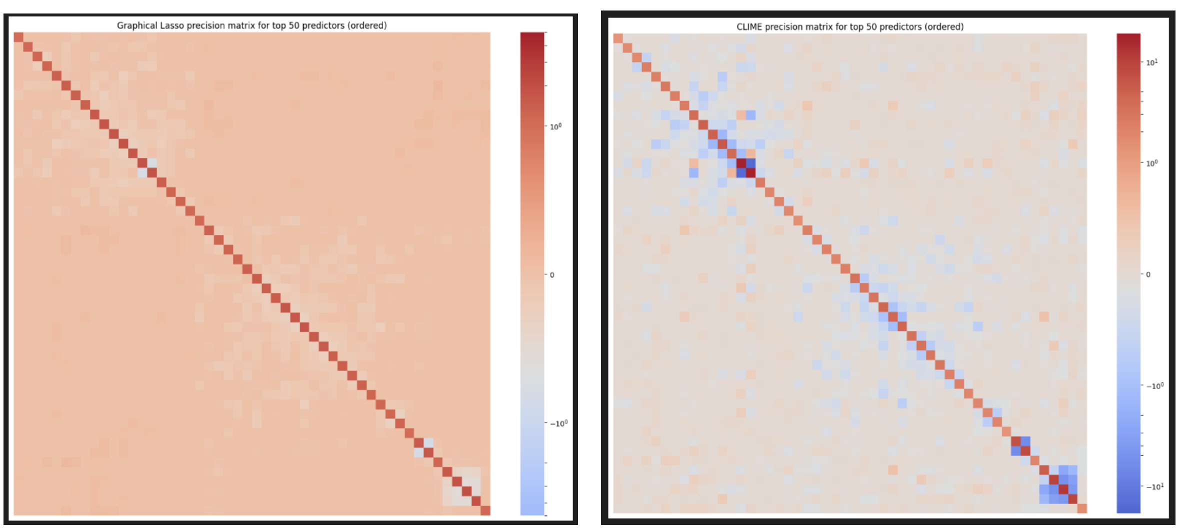

Figure 6: Estimated precision matrix by Glasso (left) and CLIME (right)

7 Conclusion

CLIME (Constrained l_1 Minimization for Inverse Matrix Estimation) is attractive largely because of its computational simplicity: it decomposes the precision matrix estimation into column-wise convex linear programs, which scale well in high dimensions compared with likelihood-based methods such as the graphical lasso (Glasso). However, there is no single method that is uniformly best across settings: the optimal choice depends on sparsity level and pattern, sample size relative to dimensionality, conditioning of the covariance, and whether the goal prioritizes support recovery (graph structure) or estimation accuracy in norm. In regimes of near or limiting sparsity where each variable has only a handful of conditional neighbors and the number of nonzeros per row grows very slowly or not at all with dimensionality, CLIME often enjoys favorable error rates and stable support recovery because its l_1-constrained formulation effectively exploits the sparse structure without incurring the heavier computational burden of likelihood optimization. By contrast, Glasso can be advantageous when the graph is moderately dense or when likelihood-based shrinkage better calibrates bias-variance tradeoffs, and nodewise regression (neighborhood selection via Lasso) has been shown in many scenarios to recover edges more accurately, especially when tuning is focused on precision in graph selection. In our breast cancer data, prior analysis indicated the precision matrix is limiting sparse, implying that conditional gene-gene dependencies form small, modular neighborhoods; this is exactly the setting in which CLIME tends to excel, which explains its strong empirical performance. Outside such sparsity regimes, there is no universal guarantee that CLIME will outperform competitors. Its primary robust advantage then is computational efficiency, and method choice should be guided by diagnostic evidence on sparsity and by the scientific objective (graph recovery vs. estimation accuracy).

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